MIL-B-11007E
21 May 1970
SUPERSEDING
MIL-B-0011007D(MU)
18 September 1968, and
MIL-B-11007B
1 June 1954

MILITARY SPECIFICATION

BURSTER, BOMB; AN-M20

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loaded burster for the AN-M47A4 100 pound PWP filled bomb.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative, Water-Repellent.

MILITARY

MIL-G-155	-	Graphite, Dry (For Use in Ammunition).
MIL-T-00339	-	Tetryl (Trinitrophenylmethylnitramine).
JAN-B-366	-	Barium Stearate.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1168	-	Lot Numbering of Ammunition.

FSC 1325

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL14-5-892 - Bomb, Burster, AN-M20, Data List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing DL14-5-892, or on the subsidiary drawings, shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.2 Assembly. The burster shall be assembled as specified on Drawing C14-5-892 and subsidiary drawings.

3.3 Pellet.

3.3.1 Density. The pellets conforming to Drawing B14-5-899, shall have a density of no less than 1.45 nor more than 1.60 grams (g) per cubic centimeter when tested as specified in 4.2.4.1.

3.3.2 Forming. The pellet, conforming to Drawing B14-5-899, shall be consolidated from tetryl conforming to grade I, class a of MIL-T-00339. Graphite conforming to grade I, II, III, or IV of MIL-G-155, barium stearate conforming to JAN-B-366, or a mixture of these materials may be uniformly blended with the tetryl as binders and lubricants to facilitate manufacture (see 6.2 and 6.3). The quantity of the binders and lubricants shall not exceed that specified on the aforementioned drawings when tested as specified in 4.2.4.2.

3.4 Workmanship. The burster shall be free of all foreign matter (oil, grease, contamination, etc) and damage such as dents, cracks, or deformation which could affect its intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of bursters, no more than one interfix number from one manufacturer of the following components is permitted.

(a) Burster casings
(b) Tetryl

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-4 for each test (pellet density and composition).

4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample bursters shall be examined in accordin accordance with the classification of defects and MIL-STD-105.

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4.2.3.2 For destructive tests. Sample pellets shall be tested in accordance with 4.2.4.1 and 4.2.4.2. The acceptance number for each test shall be zero.

4.2.3.3 Classification of defects.

(a) Burster, AN-M20 (Drawing C14-5-892).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter incorrect (Dwg B14-5-901)
102	Inside diameter incorrect (Dwg B14-5-901)
103	Camber excessive (Dwg B14-5-901)
104	Overall length incorrect
105	Component missing
106	Loose charge
107	Cement missing from thread
108	Marking incorrect
109	Workmanship (3.4)
Minor:	AQL 2.5 percent defective
201	Marking illegible
202	Protective finish missing

(b) Preparation for delivery (section 5) (Drawings D14-17-268, D14-5-2824 and D14-5-2825).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Pallets incorrect
102	Packing boxes not dried of preservative
103	Barrier liner missing (if applicable)
104	Strapping loose or missing
105	Marking incorrect
Minor:	AQL 4.0 percent defective
201	Marking illegible

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4.2.4 Tests.

4.2.4.1 Pellet density. Two calibrated aqueous zinc chloride solutions shall be prepared with density limits as specified. The pellet, when subjected to the two appropriate solutions, shall submerge in the one of lower density and float in the one of higher density.

4.2.4.2 Binders, lubricants, and inert material. A composite sample of approximately 10 g of tetryl, taken from the interior of the pellets used for the density determination, shall be weighed to the nearest 0.5 milligram in an appropriate tared extraction thimble. A Soxhlet extraction apparatus or equivalent containing approximately 50 milliliters of acetone shall be used. Heat shall be applied to the apparatus until a constant rate of reflux is reached. The minimum reflux rate shall be 60 drops per minute. The maximum reflux rate shall be such that the liquid level remains no less than 15 millimeters from the top of the thimble. The amount of solvent in the apparatus must always be sufficient to keep the tetryl in solution. The extraction shall be continued for no less than 2 hours and for 1 hour after the filtrate becomes colorless. After the extraction is completed, the thimble shall be placed in a well ventilated drying chamber, dried to a constant weight at a temperature of 50° ± 2° C, and the residue calculated as the percent of binders, lubricants, and inert material in the pellets.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. The bursters shall be packed in accordance with Drawing D14-17-268. All wood material of the box shall be preservative-treated by immersing in preservative conforming to type II, composition A, TT-W-572 for a minimum period of 3 minutes. Boxes shall not be loaded until at least 24 hours after the preservative treatment.

5.1.2 Level B. The bursters shall be packed in accordance with Drawing D14-17-268.

5.1.3 Level C. The bursters shall be packed in accordance with Drawing D14-17-268 except the waterproof barrier liner is not required.

5.2 Palletization. When specified in the contract or order (mandatory for level A), bursters, packed as specified, shall be palletized in accordance with Drawing D14-5-2825.

5.3 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified herein.

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5.3.1 Shipping container (wood box) marking. Shipping containers shall be marked in accordance with Drawing D14-5-2824. Marking shall be clear, permanent and legible after application of the wood preservative treatment.

5.3.2 Palletized load. Palletized loads shall be marked in accordance with Drawing D14-5-2824.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification, and level of packing required.

6.2 Caution. Due to the hazards inherent in the handling of explosives, loading installations are advised to take all possible precautions to protect personnel from injury while mixing and forming the explosives and loading same into the bursters.

6.3 Percent of graphite. The use of a minimum of 0.25 percent of graphite has been found to be satisfactory when forming the pellets. This information is furnished as an aid, and not as a criterion for forming.

Custodians:

Army - MU

Review activity:

Air Force - 70

Preparing activity:

Army - MU(EA)

Project No. 1325-0130

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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